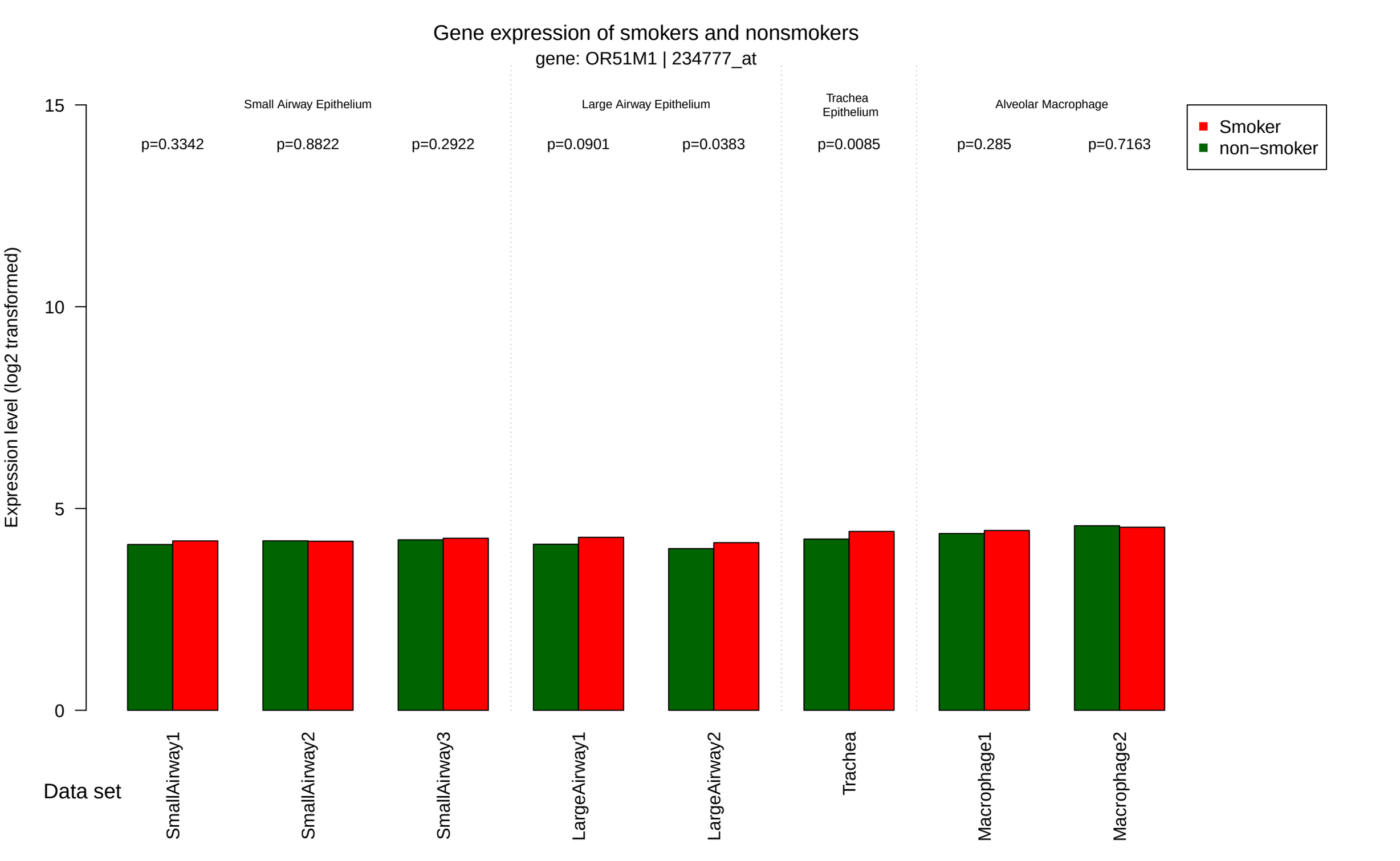
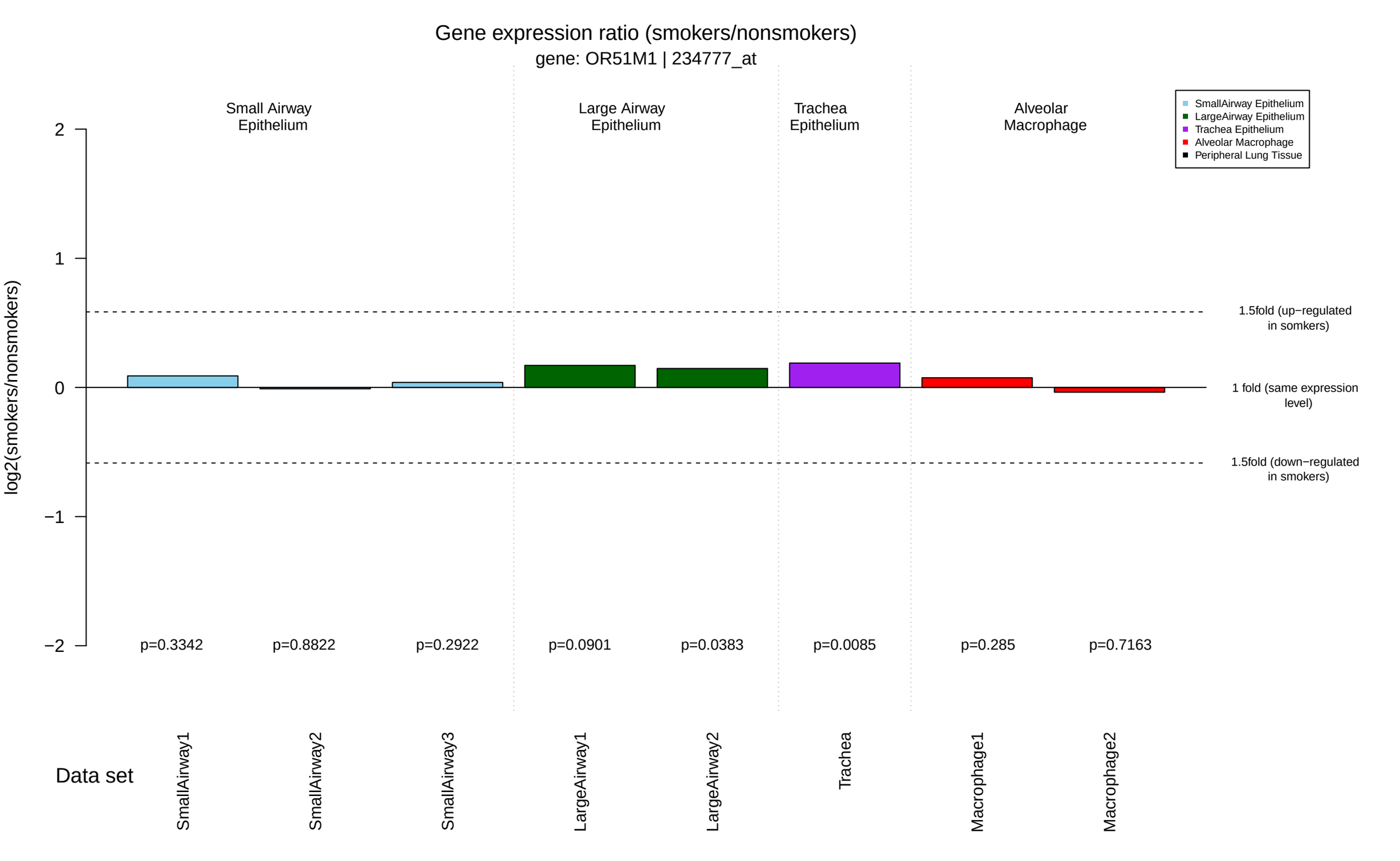
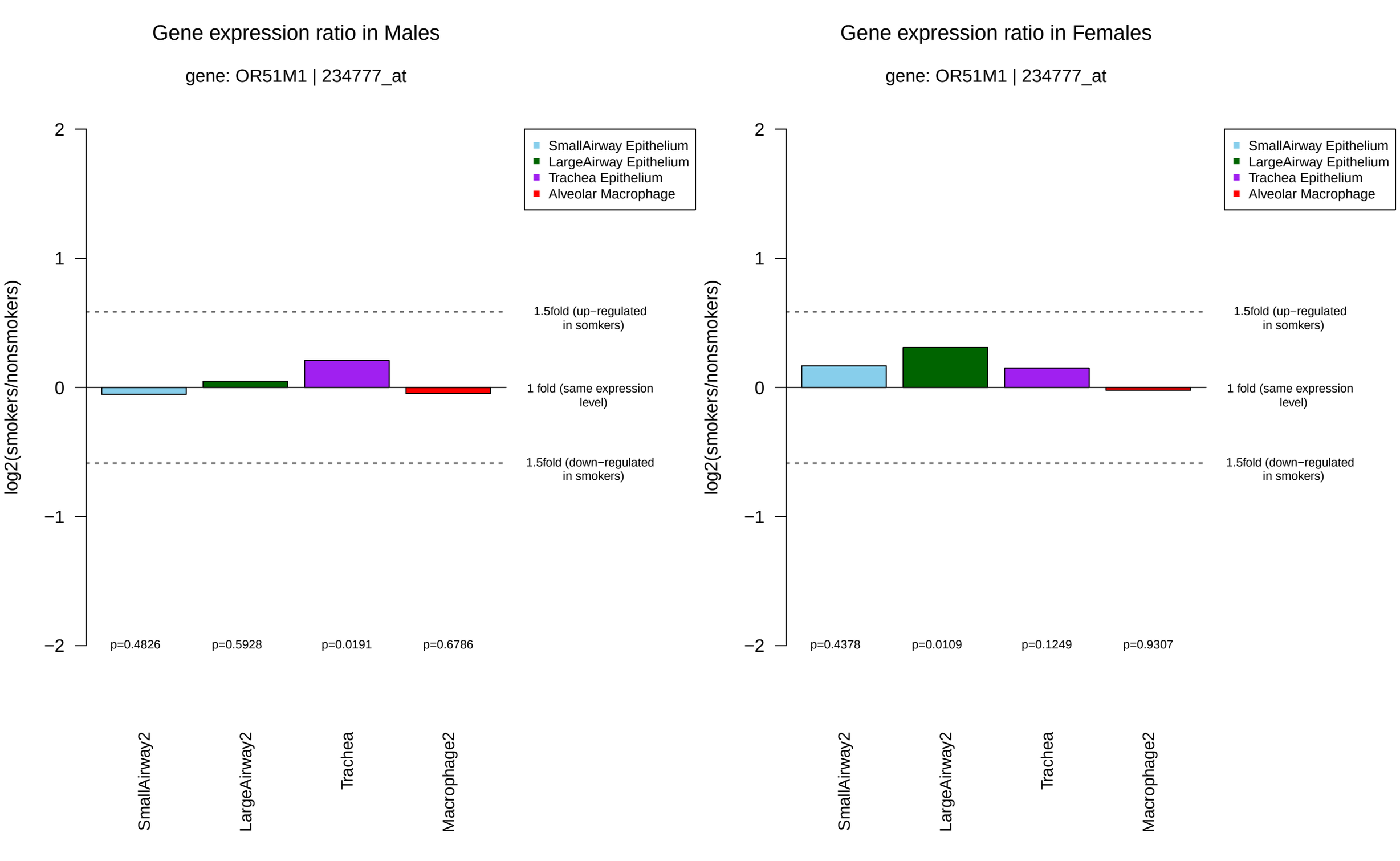
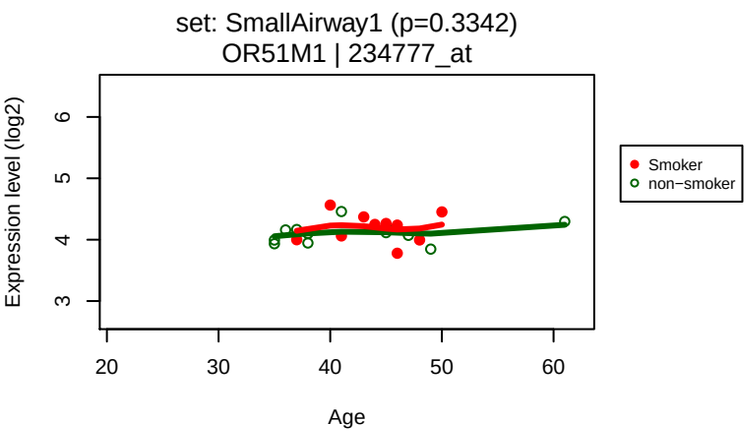
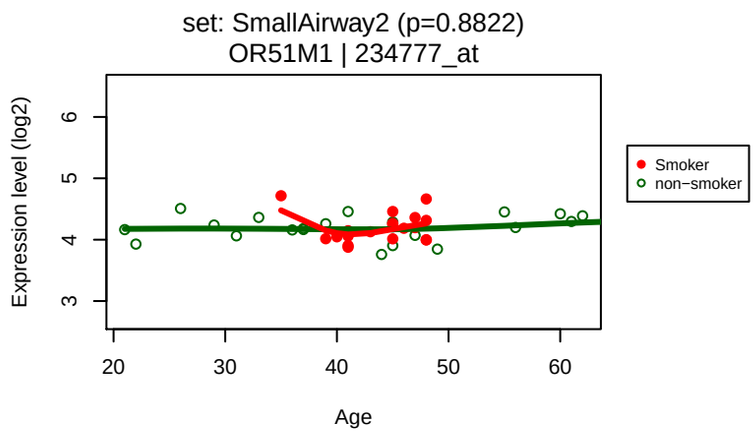
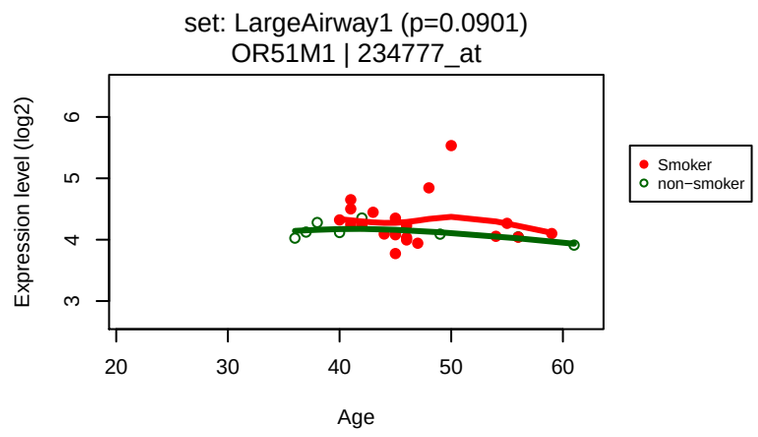

Gene expression of smokers and nonsmokers
gene: OR51M1 | 234777_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3342
p=0.8822
p=0.2922
p=0.0901
p=0.0383
p=0.0085
p=0.285
p=0.7163
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

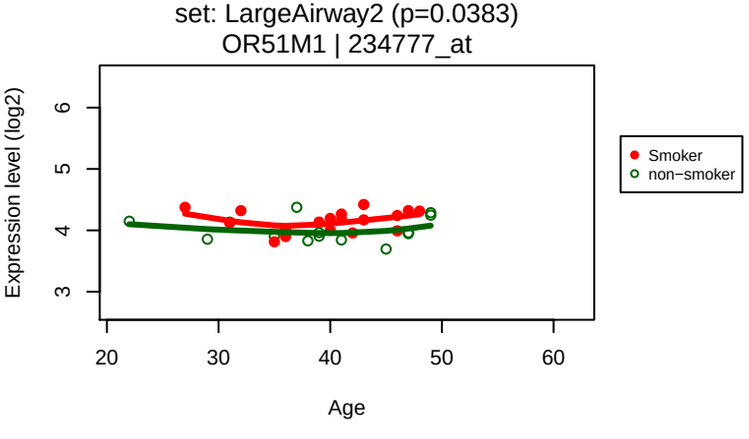

Gene expression ratio (smokers/nonsmokers)
gene: OR51M1 | 234777_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3342
p=0.8822
p=0.2922
p=0.0901
p=0.0383
p=0.0085
p=0.285
p=0.7163
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

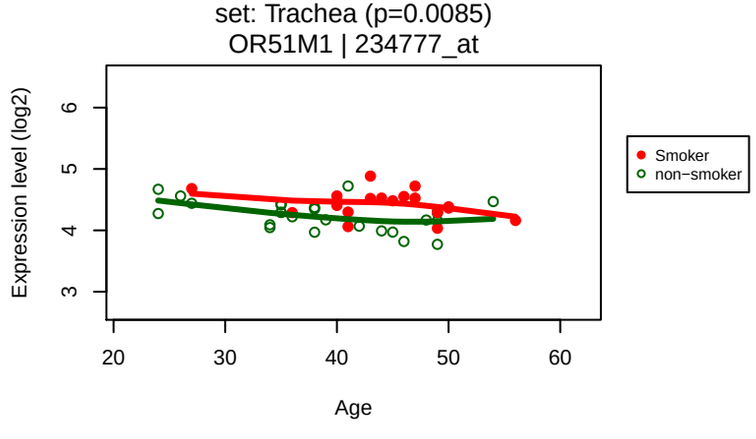

Gene expression ratio in Males
gene: OR51M1 | 234777_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4826
p=0.5928
p=0.0191
p=0.6786
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: OR51M1 | 234777_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4378
p=0.0109
p=0.1249
p=0.9307
SmallAirway2
LargeAirway2
Trachea
Macrophage2

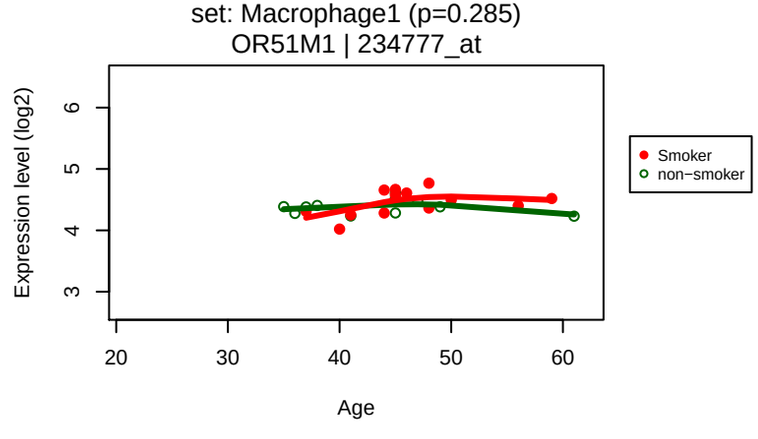

set: SmallAirway1 (p=0.3342)
OR51M1 | 234777_at
Expression level (log2)
Age
Smoker
non−smoker

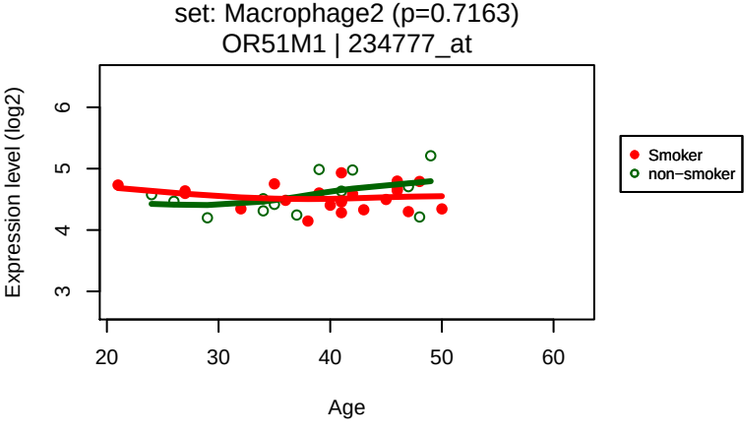

set: SmallAirway2 (p=0.8822)
OR51M1 | 234777_at
Expression level (log2)
Age
Smoker
non−smoker

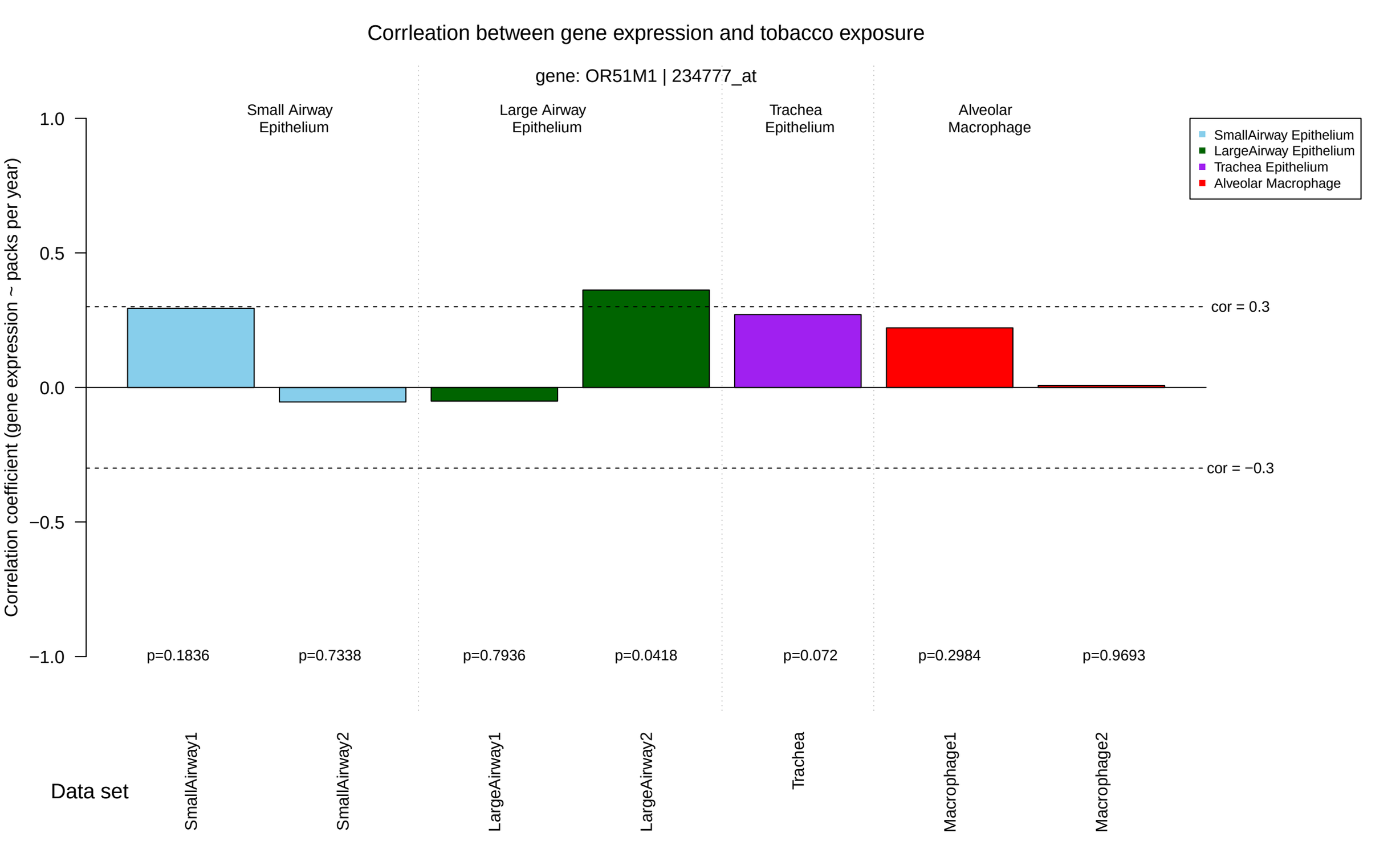

set: LargeAirway1 (p=0.0901)
OR51M1 | 234777_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0383)
OR51M1 | 234777_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0085)
OR51M1 | 234777_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.285)
OR51M1 | 234777_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7163)
OR51M1 | 234777_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: OR51M1 | 234777_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.1836
p=0.7338
p=0.7936
p=0.0418
p=0.072
p=0.2984
p=0.9693
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2